Prithvi Raj

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Education

Indian Institute of Information Technology, Guwahati (IIITG) Aug 2023 – Present
B.Tech in Electronics and Communication Engineering **CPI:** 7.73 (**SPI:** 8.71)

Semiconductor Fabrication & Research Experience

Semiconductor Workforce Development Trainee | *TCFST & NTHU, Taiwan* March 2026

- Executed hands-on, multi-layer silicon wafer fabrication in a Class 100/1000 cleanroom at National Tsing Hua University (NTHU), consistently maintaining high process stability and achieving over **90% yield** for critical layers at **2-μm precision**.
- Formulated, tested, and standardized **4 new Photolithography process recipes**, directly reducing chemical usage and substrate production waste by **30%**; techniques were adopted as standard operating procedures (SOPs) across collaboration teams.
- Practiced rigorous cleanroom protocol, mask alignment, photolithography optimization, wet/dry etching, and wafer diagnostics in line with semiconductor manufacturing standards.

Research Intern (SPARC Program) | *IIT Hyderabad & IIIT Guwahati* Nov 2025

- Conducted advanced semiconductor device fabrication research under the mentorship of Prof. Siva Krishna Vanjari.
- Successfully fabricated functional multi-layer **Resistive Gas Sensors** with **10-μm resolution**, optimizing thin-film layer deposition and thermal processing parameters for targeted Ammonia (NH_3) detection.
- Performed electrical and physical characterization of fabricated sensor nodes to identify defect margins and enhance micro-scale sensitivity.

Technical Skills

Fabrication & Cleanroom: Photolithography Process Development, Cleanroom Operations, Mask Alignment, Etching, Silicon Wafer Fabrication (2 μm & 10 μm), Yield Optimization, Thin-Film Deposition, Defect Analysis.
Device & Characterization: Resistive Gas Sensors, Semiconductor Layer Diagnostics, Microcontroller & Sensor Integration.
Programming & Simulation: C++, Python, Go (Golang), MATLAB, SQL, PyTorch, Linux/ Bash.

Applied Engineering & Systems Projects

Multi-agent Swarm Dynamics using FIREBIRD VI Jan 2026 – May 2026

Under the guidance of Dr. Surajit Panja, Associate Professor, IIIT Guwahati

- Engineered a centralized control framework using **ROS2** and **Python** to orchestrate coordinated movement across multiple agents, eliminating the need for a central server.
- Implemented graph-theoretic consensus algorithms for autonomous collision avoidance and leader-follower formations, achieving 95% success rate for 4 agents.

Photorealistic 3D Scene Reconstruction using 3D Gaussian Splatting Oct 2025 – Nov 2025

- Built a 3DGS pipeline to reconstruct traversable 3D scenes from 2D video, creating physics-consistent environments for robot path-planning simulations.
- Optimized the reconstruction process to achieve high-quality visual fidelity with lower computational overhead compared to traditional NeRFs.

Achievements & Leadership

National Finalist (Top 0.2%), India International Science Festival (IISF) S&T Hackathon 2024

- Selected among the top finalist teams out of 45,000+ national competing entries for developing hardware-software integration solutions.

1st Place Winner, GitHub University Blind Coding Challenge 2023

- Ranked #1 across all first-year engineering disciplines in a university-wide algorithmic contest.

Club Coordinator, Alankrit Fine-Arts Club – IIIT Guwahati Dec 2024 – July 2025